Classification of Riemann Surfaces

Reference: GTM81 Riemann surfaces, Forster; Complex Analysis, Stein.
Building upon our previous classification of closed surfaces, in this talk, we explore the classification of Riemann surfaces through two foundational theorems, both first developed by Riemann:

- **Uniformization Theorem**: (Analytical) Every simply connected Riemann surface is biholomorphic to either the complex sphere $\mathbb{P}^1$, the complex plane $\mathbb{C}$, or the unit disk $\mathbb{D}$.
- **Projective Embedding**: (Algebraic) Every compact Riemann surface can be embedded into projective space. Furthermore, the category of compact Riemann surfaces, smooth algebraic curves, and function fields of transcendental degree 1 are equivalent.

Preliminaries

A Riemann surface X is a (connected) topological manifold, such that for every pair of of charts $(\varphi_\alpha, U_\alpha), (\varphi_\beta, U_\beta)$, $\varphi_\alpha\varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$ is holomorphic. A function $f : X \to Y$ is holomorphic if for every local chart $(\varphi_\alpha, U_\alpha) \subset X, (\varphi_\beta, U_\beta) \subset Y$, $\varphi_\alpha f \varphi_\beta^{-1}$ is holomorphic in the usual sense.
Examples of compact Riemann surfaces:

- The Riemann surface of genus $g = 0$ is the sphere $\mathbb{P}^1$.
- Riemann surfaces of genus $g = 1$ are complex tori $\mathbb{C}/\Gamma$ where $\Gamma = \omega_1\mathbb{Z} + \omega_2\mathbb{Z}$, characterized by $\mathbb{H}/\mathrm{SL}(2, \mathbb{Z})$, or the $j-$invariant $j = 1728\frac{4a^3}{4a^3+27b^2}$ for $y^2 = x^3 + ax + b$.
  $f : X \to \mathbb{C}$ is meromorphic if $f : X \to \mathbb{P}^1$ is meromorphic. $f : X \to Y$ is biholomorphic if it has a holomorphic inverse.
  A holomorphic/meromorphic 1-form ω is a 1-form that is locally in the form $\omega = fdz$ where f holomorphic/meromorphic. Let $\mathcal{M}(X)$ be the set of meromorphic functions $f : X \to \mathbb{C}$, and $\mathcal{M}^{(1)}(X)$ meromorphic 1-forms.

Projective Embedding

Suppose X is a compact Riemann surface with genus g.
Definition: A **divisor** on X is a mapping $D : X \to \mathbb{Z}$ such that for any compact $K \subset X$, $D(x) \neq 0$ for only finite $x \in K$. (Here, its just $D(x) \neq 0$ for finite $x \in X$.) The set of divisors form an Abelian group $\mathrm{Div}(X)$, partially ordered by $D \leqslant D' \iff D(x) \leqslant D'(x) \forall x \in X$.
Definition: For open $Y \subset X$, $f \in \mathcal{M}(Y)$ and $a \in Y$, define

$$\mathrm{ord}_a(f) = \begin{cases} 0, & f \text{ holomorphic and nonzero at } a \\ k, & f \text{ has a zero of order } k \\ -k, & f \text{ has a pole of order } k \\ \infty, & f \text{ vanishes locally} \end{cases}$$

Then $x \mapsto \mathrm{ord}_x(f)$ is a divisor, denoted (f) and called a principal divisor.
Definition: If X is compact, one can define a homomorphism

$$\deg : \mathrm{Div}(X) \to \mathbb{Z}, D \mapsto \sum_{x \in X} D(x).$$

Then $\deg(f) = 0$ for any $f \in \mathcal{M}(X)$. For a meromorphic 1-form $\omega = df$ and $a \in X$, suppose locally $\omega = gdz$, then we define $\mathrm{ord}_a(\omega) = \mathrm{ord}_a(g)$. Since every pair of 1-forms ω_1, ω_2 satisfy $\omega_1 = f\omega_2$, $(\omega_1) = (\omega_2) + (f)$, so the equivalence class $[K] = [(\omega)]$ is independent of the choice of ω.

Theorem (Riemann-Roch): Suppose D is a divisor on a compact Riemann surface X of genus g, then $H^0(X, \mathcal{O}_D)$ and $H^1(X, \mathcal{O}_D)$ are finite dimensional vector spaces, and

$$\mathrm{dim} H^0(X, \mathcal{O}_D) - \mathrm{dim} H^1(X, \mathcal{O}_D) = 1 - g + \deg D.$$

Here, $H^0(X, \mathcal{O}_D) = L(D) = \{f \in \mathcal{M}(X) : (f) \geqslant -D\}$.
By Serre duality, we know

$$\mathrm{dim} H^1(X, \mathcal{O}_D) = \mathrm{dim} H^0(X, \mathcal{O}_{K-D}),$$

hence Riemann-Roch can be written as

$$\mathrm{dim} L(D) - \mathrm{dim} L(K - D) = 1 - g + \deg D.$$

As a corollary, let $D = 0, K$ we obtain $\deg K = 2g - 2$. This also gives another definition of genus: $g = \mathrm{dim} L(K)$.
Using Riemann-Roch, we can **classify compact Riemann surfaces of genus zero**:
Suppose X is a compact Riemann surface of genus g, and $a \in X$, then set $D(a) = g + 1$ and $D(x) = 0$ otherwise, we obtain $\mathrm{dim} L(D) \geqslant 1 - g + \deg D = 2$ so there is a non-constant $f \in L(D)$. Then f is meromorphic, with a pole of order $\leqslant g + 1$ at a, and otherwise holomorphic, so f is a holomorphic covering map $f : X \to \mathbb{P}^1$ with at most $g + 1$ sheets. When $g = 0$, f is a biholomorphism so X is isomorphic to $\mathbb{P}^1$.

Let D be a divisor on X, we call D (the sheaf $\mathcal{O}_D$) globally generated if for any x there exists $f \in L(D)$ such that $\mathrm{ord}_x(f) = -D(x)$.
Theorem: If X has genus g, and $\deg D \geqslant 2g$, then D is globally generated.
Proof: Suppose $x \in X$ is a fixed point, let $D' = D - x$ ($D'(y) = D(y) \forall y \neq x$ and $D'(x) = D(x) - 1$). Since $\deg D > \deg D' > 2g - 2$, $\deg(K - D') < 0$ so we have $L(K - D) = L(K - D') = 0$, and by Riemann-Roch $\mathrm{dim} L(D) > \mathrm{dim} L(D')$. Take $f \in L(D) \backslash L(D')$, then $\mathrm{ord}_x(f) = -D(x)$.

For a compact surface X of genus $g > 0$, let D be a divisor of degree $\geqslant 2g + 1$. Take a basis $f_0, \cdots, f_N$ of $L(D)$, we construct an embedding $F = (f_0 : \cdots : f_N) : X \to \mathbb{P}^N$:
(i) Definition: For any $x \in X$, take a neighborhood (V, z) where $z(x) = 0$, and let $k = \min \mathrm{ord}_x(f_i)$. Suppose $f_i(z) = z^k g_i(z)$ then g_i are holomorphic and $g_i(x)$ not identically zero, and we can define $F(x) = (g_0(x) : \cdots : g_N(x))$. Suppose $g_j(x) \neq 0$, then locally F can be viewed as $F : X \to \mathbb{C}^N, F = \left(\frac{g_0}{g_j}, \cdots, \frac{g_{j-1}}{g_j}, \frac{g_{j+1}}{g_j}, \cdots, \frac{g_N}{g_j}\right)$, so F is holomorphic.
(ii) Injective: Suppose $x_1 \neq x_2$ in X, take $D' = D - x_2$, then $\deg D' = 2g$ so D' is globally generated. Take $f \in L(D')$ such that $\mathrm{ord}_{x_1}(f) = -D(x_1)$, and $\mathrm{ord}_{x_2}(f) > -D(x_2)$. Suppose $f = \sum \lambda_i f_i$, and take local coordinates $(V_1, z_1), (V_2, z_2)$ such that $z_i(x_i) = 0$. Suppose $k_j = \min \mathrm{ord}_{x_j}(f_i)$, and $f_i(z_j) = z_j^{k_j} g_{i,j}(z)$. Then by construction of f, $g_1(x_1) \neq 0$ while $g_2(x_2) = 0$, so $F(x_1) \neq F(x_2)$ and F is injective.
(iii) Immersion: Let $x_0 \in X$ and $D' = D - x_0$. Take $f \in L(D')$ such that $\mathrm{ord}_{x_0}(f) = -D(x_0) + 1$. Suppose

$f = \sum \lambda_i f_i$, and let (V, z) be a neighborhood such that $f = z^k g, f_i = z^k g_i$ where $k = \min \operatorname{ord}_{x_0}(f) = -D(x_0)$. WLOG assume $g_0(x_0) \neq 0$. Then $F_0 = \varphi_0 \circ f = (F_{01}, \cdots, F_{0N}) : V \to \mathbb{C}^N$ is given by $F_0 = \left(\frac{g_1}{g_0}, \cdots, \frac{g_N}{g_0}\right)$, and $\sum \lambda_i dF_{0i} = d(g/g_0)$. Since $g_0(x_0) \neq 0$ and g has a simple zero at x_0, $d(g/g_0) \neq 0$ so at least one $dF_{0i} \neq 0$.

Recall that given homogeneous polynomials $\{F_\alpha\}$, an algebraic variety $C \subset \mathbb{P}^n$ is $\{p : F_\alpha(p) = 0 \forall \alpha\}$. Algebraic curves are varieties of dimension 1, and an analytic variety is locally the zero set of a holomorphic function.
Chow's Theorem: Any analytic subvariety of projective space is algebraic. Holomorphic maps between analytic subvarieties induce regular rational morphisms between algebraic varieties.
Therefore, every compact Riemann surface is isomorphic to a smooth algebraic curve.

Recall the classic theorem that a topological space X is completely determined by $C(X)$, namely $X \cong \operatorname{Max}(C(X))$. There is a similar result for Riemann surfaces:
For a compact Riemann surface X, $\mathcal{M}(X) \cong \mathbb{C}[x, y]/P(x, y)$ is a function field of transcendental degree 1, and X is determined by the DVRs of $\mathcal{M}(X)$ containing $\mathbb{C}$. Any homomorphism $f : X \to Y$ gives a $\mathbb{C}-$algebra hom $f^* : \mathcal{M}(Y) \to \mathcal{M}(X)$.

Therefore, we obtain a three-fold categorical equivalence:
Compact Riemann surfaces - Smooth algebraic curves - Function fields over $\mathbb{C}$ with degree 1.
And objects like point-DVR or Deck transformation-Galois group correspond to each other. Hence we can study the same objects using tools from both analysis and algebra.

Uniformization Theorem

Main Theorem: If X is a simply connected Riemann surface, then X can be biholomorphically mapped onto either the Riemann sphere $\mathbb{P}^1$, the complex plane $\mathbb{C}$, or the unit disk $\mathbb{D}$.

Recall the proof of Riemann mapping theorem, we first show simply connected open sets are biholomorphic to $\mathbb{D}$, then solve the Dirichlet problem using the biholomorphism. Here, the process is reversed.

Lemma1: Suppose X is a non-compact Riemann surface, and $Y \subset X$ is simply connected with ∂Y piece-wise smooth. Then there exists a biholomorphic mapping of Y onto the unit disk $\mathbb{D}$.
(This is PDE, using harmonic analysis and Weyl's lemma to solve the Dirichlet problem.)
Koebe-Bieberach Theorem: If $f : \mathbb{D} \to \mathbb{C}$ holomorphic and injective, $f(0) = 0, f'(0) = 1$, then $D\left(\frac{1}{4}\right) \subset f(\mathbb{D})$, and $|f(z)| \leqslant |z|/(1 - |z|)^2$. See Stein, Problem3.1.

Definition: A family $\mathcal{F}$ of holomorphic functions is called normal if every sequence in $\mathcal{F}$ has subsequence converging uniformly on every compact subset (the limit may not be in $\mathcal{F}$).
Montel's theorem: Recall Arzelá-Ascoli, locally uniform boundedness & equicontinuous implies $\mathcal{F}$ is a normal family. By the Cauchy integral formula uniform boundedness implies equicontinuous.
Hurwitz's theorem: Note that if f_n is injective and converges uniformly to f on every compact subset, then f is injective or constant, using the Cauchy integral formula.

Proof of main theorem:
(i) If X is compact, then X is of genus $g=0$ since it is an orientable & simply connected closed surface (using classification of closed surfaces), so X is the Riemann sphere $\mathbb{P}^1$.
(ii) Now assume X is not compact.
Take an exhaustion of connected compact sets $K_0 \subset K_1 \subset \cdots$. For each n, find a relatively compact open neighborhood $V_n \supset K_n$ such that ∂V_n is piece-wise smooth (union of coordinate disks). Let Y_n be the union of V_n and compact components of $X \backslash V_n$. Since X is simply connected, so are Y_n hence by Lemma1 each Y_n is biholomorphic to $\mathbb{D}$.
Choose a point $a \in Y_0$ and a coordinate neighborhood (U, z) of a, then there exists a real number $r_n > 0$ and a biholomorphic mapping $f_n : Y_n \to D(r_n)$ with $f_n(a) = 0$ and $f_n'(a) = 1$.
(iii) Note that $r_n \leqslant r_{n+1}$ for all n, since $h = f_{n+1} \circ f_n^{-1} : D(r_n) \to D(r_{n+1})$ satisfies $h(0) = 0$ and $h'(0) = 1$, so by Schwarz lemma $1 \leqslant r_{n+1}/r_n$. Let $R = \lim_{n\to\infty} r_n \in (0, \infty]$. We show that X is biholomorphic to $D(R)$ which is either $\mathbb{D}$ or $\mathbb{C}$.
Let $h_{m,n} = f_m f_n^{-1}$ then its normalization $g(x) = h_{m,n}(xr_n)/r_n$ is injective and $g(0) = 0, g'(0) = 1$.
(iv) We show that $\{f_n\}$ is normal, and obtain $f : X \to D(R)$.
Case1: $R < \infty$, then f_n is bounded by R.
Case2: $R = \infty$. By Koebe distortion theorem $|g(z)| \leqslant |z|/(1-|z|)^2$, so for $w \in D(r_n)$, $|h_{m,n}(\omega)| \leqslant r_n^2|\omega|/(r_n - |\omega|)^2$. So $h_{m,n}$ and also f_n are locally uniformly bounded.
By Montel's theorem $\{f_n\}$ has a subsequence that converges uniformly on any compact subset, and denote the limit by f. Since f_n are injective, f is injective, $f(a) = 0$ and $f'(a) = 1$.
(v) Finally we show that $f : X \to D(R)$ is surjective:
Case1: $R = \infty$. Consider $g : \mathbb{D} \to \mathbb{C}$ of $h_{m,n}$, then by Koebe 1/4 theorem $D(1/4) \subset g(\mathbb{D})$. Hence $D(r_n/4) \subset h_{m,n}(D(r_n)) = f_m(Y_n)$. Let $m_k \to \infty$ we obtain $f(X) \supset f(Y_n) \supset D(r_n/4)$, so $f(X) \supset \bigcup_{n=1}^{\infty} D(r_n/4) = \mathbb{C}$. Therefore $f(X) = \mathbb{C}$.
Case2: $R < \infty$. Suppose the contrary, we find a biholomorphic map $g : f(X) \to D(r)$ with $r < R$ such that $g(0) = 0$ and $g'(0) = 1$. Take $r_n > r$, then $h = g \circ f \circ f_n^{-1} : D(r_n) \to D(r)$ satisfy $h(0) = 0, h'(0) = 1$ so it contradicts Schwarz lemma.
Let $Y = f(X)$ be a proper simply connected subset of $D(R)$, we can suppose $R = 1$. Take $a \in \mathbb{D} \backslash Y$, and $\psi_a = \frac{z-a}{1-z\bar{a}} \in \mathrm{Aut}(\mathbb{D})$. Since $\psi_a|_Y \neq 0$, take $g : Y \to \mathbb{D}$ such that $g^2 = \psi_a|_Y$. Let $b = g(0)$ and take ψ_b, then $h = \psi_b \circ g$ satisfy $h(0) = 0, h'(0) = \gamma = (1+|b|^2)/2b$. Hence $h/\gamma : Y \to D(1/|\gamma|)$ has the desired properties.
(Same as Stein's proof of the Riemann mapping theorem.)

Applications: For a Riemann surface X, if the universal covering $\tilde{X}$ is $\mathbb{P}^1/\mathbb{C}/\mathbb{D}$, we call it elliptic/parabolic/hyperbolic.
Recall that X can be viewed as $\tilde{X}/G$ where $G = \mathrm{Deck}(\tilde{X}/X)$ acts freely and discretely.
Proposition: (a) Every automorphism of $\mathbb{P}^1$ has a fixed point.
(b) If $G < \mathrm{Aut}(\mathbb{C})$ which acts discretely and freely, $G = \langle id \rangle$, $G = \langle \tau_\gamma \rangle$ where $\tau_\gamma : z \mapsto z + \gamma$, or $G = \langle \tau_{\gamma_1}, \tau_{\gamma_2} \rangle$ where γ_1, γ_2 are independent over $\mathbb{R}$.
Easy to verify since automorphism of $\mathbb{P}^1$ and $\mathbb{C}$ are in the form $\frac{az+b}{cz+d}$ and $az+b$.
Corollary: (a) $\mathbb{P}^1$ is elliptic; (b) $\mathbb{C}, \mathbb{C}^*, \mathbb{C}/\Gamma$ are parabolic; (c) all other Riemann surfaces are hyperbolic.
We obtain Picard's little theorem as an easy corollary:
Corollary: Suppose $f : \mathbb{C} \to \mathbb{C}$ is non-constant and entire, then f takes every value of $\mathbb{C}$ with at most one exception.